**Abbreviations**
cao – correct answer only
dep – dependent
FT – follow through after error
isw – ignore subsequent working
oe – or equivalent
SC – Special Case
nfww – not from wrong working
soi – seen or implied

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | 2 840 327 | **1** | |
| 2 | Obtuse | **1** | |
| 3(a) | 77 | **1** | |
| 3(b) | Ruled perpendicular line meeting the given line | **1** | |
| 4 | Correct triangle with correct arcs | **2** | **B1** for correct triangle with no arcs or incorrect arcs or two correct arcs<br><br>If 0 scored, **SC1** for correct triangle and correct arcs but lines interchanged |
| 5 | Any multiple of 72 | **1** | |
| 6 | $\frac{4}{25}$ cao | **2** | **M1** for $\frac{32}{200}$ oe |
| 7(a) | Oslo | **1** | |
| 7(b) | 9 | **1** | |
| 8 | 54 | **2** | **M1** for 180 – 71 – 55 oe or **B1** for 55 or 125 in a relevant correct position on the diagram |
| 9 | 140, 60 | **2** | **M1** for $\frac{200}{(7+3)} \times k$ where $k$ = 1, 7 or 3 |
| 10(a) | 6 7<br>1 2 6 7<br>1 3 7<br>0 2 | **2** | **B1** for 2 rows correctly ordered or fully correct unordered |
| 10(b) | 2.7 cao | **1** | |
| 11(a) | $\frac{90}{360} \times 240$ oe | **1** | |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 11(b) | Ruled line giving sector as 165° | **2** | **M1** for $\frac{110}{240}\times 360$ or $\frac{90}{60}\times 110$ oe |
| 12 | 442 | **2** | **M1** for $\frac{100-15}{100}\times 520$ oe or **B1** for 78 |
| 13 | $\frac{2}{6}+\frac{5}{6}$ oe | **M1** | i.e. correct fractions with common denominator 6*k* |
| | $1\frac{1}{6}$ cao | **A1** | |
| 14(a) | 0.96 or 96% or $\frac{24}{25}$ final answer | **1** | |
| 14(b) | 34 cao | **2** | **M1** for 0.04 × 850 oe |
| 15 | 420 | **2** | **M1** for $\frac{330}{11}\times 14$ oe |
| 16(a) | –1.25, 1.25 oe | **2** | **B1** for each |
| 16(b) | Correct curve | **4** | **B3FT** for 9 or 10 correctly plotted points<br>**B2FT** for 7 or 8 correctly plotted points<br>**B1FT** for 5 or 6 correctly plotted points |
| 17(a) | × 3 | **1** | |
| 17(b) | $4n + 9$ oe final answer | **2** | **B1** for $4n + k$ or $jn + 9$, $j \neq 0$, or correct answer seen then spoilt |
| 18(a) | *a*, *b*, *c*, *d* | **1** | |
| 18(b) | 6 | **1** | |
| 19 | 317 | **2** | **M1** for 137 + 180 oe |
| 20(a) | $2.73 \times 10^{-3}$ cao | **1** | |
| 20(b) | 42.1 is incorrectly rounded [to 3 significant figures] oe<br><br>$42.1 \times 10^{6}$ is not correct standard form oe | **2** | **B1** for each<br><br>If **B0** scored, **SC1** for $4.22\times 10^{7}$ seen |
| 21 | 34 | **2** | **M1** for $\frac{BC}{4}=\frac{51}{6}$ oe or better |